CBAM Spillovers and the Industrial Design of Carbon Pricing Instruments

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March 28, 2026

Abstract

Does the EU Carbon Border Adjustment Mechanism shape global carbon pricing adoption and the sectoral design of new instruments? Using a World Bank State & Trends panel (161 CPIs, 2016–2026), BACI bilateral trade data (196 countries, 2014–2024), and WDI/WGI controls, we find three headline results. First, CBAM-exposed countries advance faster through the carbon pricing pipeline: Cox hazard ratios for the development transition reach 1.20 ($p < 0.01$). Second, the 2019 cohort of new instruments shows markedly higher CBAM exposure (mean 2.1%, ten times the 2016–2018 average) and a 100% industry coverage rate. Third, panel difference-in-differences coefficients for industry coverage are consistently positive but imprecisely estimated given 49 country clusters. These results are directionally consistent with CBAM driving industry-oriented instrument design, motivating continued panel extension as the mechanism transitions to full implementation.

JEL codes: Q54, Q58, F18, H23

1 Introduction

The EU Carbon Border Adjustment Mechanism creates a direct financial incentive for non-EU countries to establish domestic carbon pricing. Under Article 9 of Regulation (EU) 2023/956, importers may deduct from their CBAM certificate obligations any carbon price already paid in the country of origin (European Parliament and Council, 2023). This rebate mechanism means that exporting countries without domestic carbon pricing effectively transfer carbon tax revenues to the EU treasury. Whether this incentive has accelerated both carbon pricing adoption and specifically industry-oriented instrument design is the central empirical question of this paper. The EU accounts for roughly 6% of global emissions; the CBAM mechanism matters for climate policy only if it catalyses broader policy diffusion beyond Europe's borders.

We find three headline results. First, countries with greater CBAM trade exposure advance faster through the carbon pricing policy pipeline: Cox proportional hazard models yield a hazard ratio of 1.201 ($p < 0.01$) for the development transition, meaning a one percentage point increase in CBAM exposure is associated with a 20% higher hazard of moving into active carbon pricing development (Table 5). Second, the 2019 entrant cohort—twelve new instruments entering the World Bank database in the year the European Commission first announced detailed CBAM plans—shows a mean CBAM exposure of 2.1%, ten times higher than the 2016–2018 cohorts, with 100% industry sector coverage from inception (Table 7). Third, CPI-level panel difference-in-differences estimates for industry coverage are consistently positive (coefficients of +0.97 to +1.11) but imprecisely estimated with 49 country clusters (Table 9).

Our work extends the framework of Bahí et al. (2026), who establish that more CBAM-exposed countries were more likely to adopt some form of national carbon pricing after 2019, using a country-year two-way fixed effects (TWFE) design with a binary adoption outcome. We build on their contribution in two directions. First, we test whether the policy response is visible across the full pipeline from consideration through development to implementation using event-history models, moving beyond the binary adoption indicator. Second, we introduce a sectoral dimension—testing whether new instruments in more-exposed jurisdictions are more likely to cover CBAM-relevant industrial sectors from inception. The binary country-level outcome in Bahí et al. (2026) cannot distinguish between a country that adopts a power-sector-only carbon tax and one that adopts an industry-covering ETS specifically in response to CBAM incentives.

The remainder of this paper is organised as follows. Section 2 describes the data and measurement. Section 3 presents the empirical strategy. Section 4 reports results by analytical layer. Section 5 presents robustness checks. Section 6 concludes.

2 Data and Measurement

2.1 Carbon Pricing Instrument Panel

Our primary data source is the World Bank *State and Trends of Carbon Pricing* data series (World Bank, 2025a). We construct a panel covering 161 carbon pricing instruments (CPIs) across 11 annual editions (S&T16 through S&T26, publication years 2016–2026). For each CPI and year we record nine binary sector coverage indicators: power, industry, road transport, aviation, shipping, buildings, waste, agriculture, and forestry. These are parsed from comma-separated sector strings in the source data. We also record time-varying policy status anchored to master-list dates: under consideration start, under development start, and implementation start. These master-list dates differ from first database appearance for some instruments—we document a median gap of 0 years between master-list development start and first sectoral coverage database entry, with Japan's national ETS showing a three-year gap as a concrete example. Annual carbon prices are recorded in nominal USD, converted at April 1 exchange rates. Each CPI is classified by the host country's income group (World Bank) and World Bank region.

Several descriptive facts motivate our empirical design. Industry coverage grew from 36 CPIs in 2016 to 79 in 2026—the largest absolute gain of any sector. Of the 52 CPIs that first covered industry after 2016, 51 entered the coverage database already covering industry from inception. Only China's national ETS (ETS_CN) genuinely expanded to cover industry after its initial launch—from power only in 2021 to power and industry in 2023. The 2019 entrant cohort stands out: twelve new instruments with mean CBAM exposure of 2.1% (ten times higher than 2016–2018 cohorts) and 100% industry coverage at entry.

Table 1: Sector Coverage Counts by Year

Sector	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Power	31	36	38	46	51	54	53	58	59	59	62
Industry	36	40	41	51	55	59	67	72	72	75	79
Road transport	19	23	24	30	31	34	36	38	35	32	32
Aviation	14	16	17	21	23	23	24	26	23	21	22
Shipping	8	8	9	14	16	19	18	20	18	17	18
Buildings	26	30	31	35	36	38	42	45	45	46	47
Waste	6	6	6	7	7	7	7	8	10	15	16
Agriculture	12	14	15	18	19	20	25	26	28	31	31
Forestry	3	3	3	3	3	3	3	3	3	3	3

Note: Count of CPIs with active sector coverage in each year. Source: World Bank State & Trends of Carbon Pricing panel, 2016–2026.

2.2 CBAM Trade Exposure

Our CBAM exposure variable measures the share of a country's total merchandise exports that flow to EU27 plus UK destinations in CBAM-covered HS6 product categories. We construct this from BACI HS12 bilateral trade data (Gaulier and Zignago, 2010) covering 196 non-EU, non-EFTA exporting countries over 2014–2024.

CBAM-covered product categories follow Annex I of Regulation (EU) 2023/956 (European Parliament and Council, 2023): iron and steel (HS chapters 72–73, selected codes), aluminium (HS chapter 76), cement (HS 2523), fertilisers (HS chapter 31), electricity (HS 271600), and hydrogen (HS 280410).

The importer perimeter is time-varying: EU27 only for 2014–2019, EU27 plus UK from 2020 onward, reflecting Brexit and the UK's subsequent announcement of its own CBAM. We exclude EU27 and EFTA countries from the exporter sample since they are not subject to CBAM as exporters.

Our benchmark exposure measure follows Bahí et al. (2026): a static average of annual exposure over 2014–2024 (denoted $\overline{E}_i$). We also construct a time-varying annual measure $E_{i,t}$ as our preferred specification, which better captures the Brexit correction and avoids averaging over a period with a changing importer perimeter.

2.3 Control Variables

We include GDP per capita (constant 2015 USD) and total exports from the World Development Indicators (World Bank, 2025b), and government effectiveness, regulatory quality, rule of law, and control of corruption from the Worldwide Governance Indicators (Kaufmann et al., 2025). We construct a composite WGI index as the simple mean of the four governance indicators. All controls are lagged one year to avoid simultaneity. Coverage spans 273 countries over 2015–2024.

3 Empirical Strategy

Our analysis proceeds in four layers, each addressing a distinct empirical question with progressively richer outcome measurement. Layer A tests the broad adoption question established by Bahí et al. (2026): does CBAM exposure predict any carbon pricing activity? This serves as a benchmark for comparability, not as our primary contribution. Layer B tests the pipeline question: does exposure predict faster movement from consideration through development to implementation? This is the paper's primary empirical contribution, estimated via both TWFE difference-in-differences and event-history models (Cox proportional hazards and discrete-time complementary log-log). Layer C tests the design question: conditional on entering the carbon pricing system, are new instruments in more-exposed jurisdictions designed

Table 2: Top 20 Countries by CBAM Trade Exposure

Rank	Country	ISO3	Exposure (%)
1	Montenegro	MNE	22.33
2	Mozambique	MOZ	15.41
3	Bosnia Herzegovina	BIH	9.64
4	Albania	ALB	8.15
5	Ukraine	UKR	7.67
6	Serbia	SRB	7.65
7	North Macedonia	MKD	6.92
8	New Caledonia	NCL	6.23
9	Armenia	ARM	6.22
10	Rep. of Moldova	MDA	4.81
11	Egypt	EGY	4.12
12	Bahrain	BHR	4.08
13	Belarus	BLR	3.78
14	TÃ¼rkiye	TUR	3.12
15	Tajikistan	TJK	2.79
16	Russian Federation	RUS	2.79
17	Bahamas	BHS	2.40
18	Georgia	GEO	2.38
19	Morocco	MAR	1.99
20	Turks and Caicos Isds	TCA	1.94

Note: Exposure = share of total exports to EU27+UK in CBAM-covered HS product categories, averaged 2014–2024. Bahí et al. (2026) report Montenegro 24.2% and Turkey 4% using EU+EFTA destination; our figures (22.18% and 3.08%) differ due to the EU27+UK perimeter.

to cover CBAM-relevant industrial sectors from inception? This is the paper's primary substantive contribution, though statistically constrained by sample size and limited to multi-specification directional evidence. Layer D provides secondary evidence on the price intensive margin.

3.1 Benchmark Specification (Layer A)

This layer replicates the framework of Bahí et al. (2026) for comparability and serves as a benchmark, not the paper's primary claim. Our benchmark replicates the Bruegel-comparable TWFE design. The estimating equation is:

$$Y_{c,t} = \alpha_c + \lambda_t + \beta \left(\overline{E}_c \times \mathbf{1}[t \geq \tau] \right) + \gamma' \mathbf{X}_{c,t-1} + \varepsilon_{c,t} \tag{1}$$

where $Y_{c,t}$ is a binary outcome for country c in year t, α_c are country fixed effects, λ_t are year fixed effects, $\overline{E}_c$ is the benchmark static CBAM exposure, $\mathbf{1}[t \geq \tau]$ is a post-milestone indicator for $\tau \in \{2019, 2021, 2023\}$, and $\mathbf{X}_{c,t-1}$ are lagged controls. Standard errors are clustered at the country level. The sample spans 196 non-EU/EFTA countries over 2016–2024.

The identification logic relies on CBAM milestones as common shocks to all countries; cross-sectional variation in $\overline{E}_c$ generates treatment intensity variation. Bahí et al. (2026) find $\hat{\beta} \approx 0.016$ per percentage point increase in exposure. Our exposure is measured as a proportion (0–1) rather than percentage points (0–100). To compare with Bruegel, our coefficients should be divided by 100, yielding $\hat{\beta}/100 \approx 0.013$, consistent with their estimate.

3.2 Pipeline Extension (Layer B)

We extend the TWFE specification to pipeline outcomes using master-list timing (ever considered, ever developing, ever implemented). We also estimate Cox proportional hazard models:

$$h_c(t) = h_0(t) \exp\left(\beta_1 E_c^{\text{pct}} + \beta_2 \log \text{GDP}_c + \beta_3 \text{WGI}_c \right) \tag{2}$$

where $h_c(t)$ is the hazard of country c reaching a pipeline milestone (consideration, development, implementation) at time t, E_c^{pct} is CBAM exposure in percentage points, and $h_0(t)$ is the baseline hazard. Countries already in the relevant state before 2016 are excluded via milestone-specific left-censoring.

We also estimate discrete-time complementary log-log (cloglog) models, which provide a natural link between the continuous-time proportional hazards framework and the annual panel structure of our data:

$$\log\big[-\log(1-h_{c,t})\big] = \alpha_t + \beta_1 E_{c,t} + \gamma' \mathbf{X}_{c,t-1} \tag{3}$$

where $h_{c,t}$ is the discrete-time hazard of country c transitioning at year t, α_t captures the baseline hazard via year indicators, and $E_{c,t}$ is time-varying CBAM exposure. The discrete-time specification is our preferred event-history model because it accommodates the annual data structure naturally, allows time-varying covariates including the preferred Brexit-corrected exposure measure, and avoids the tied-event approximations required by Cox models in discrete time. Cox models are retained as a robustness check; we disclose the penalizer parameter and events-per-variable ratios throughout.

3.3 Entrant-Level Sectoral Design (Layer C)

The near-absence of within-instrument sectoral expansion—51 of 52 post-2016 industry-covering CPIs entered with industry already covered—motivates an entrant-level design. The question is not whether existing instruments expand to cover industry, but whether *new* instruments in CBAM-exposed countries are more likely to cover industry from inception. We present entrant-level descriptive and regression evidence supported by a battery of four supplementary specifications with different identifying assumptions and units of analysis. Because the sample is conditioned on countries that have moved into the carbon pricing policy set, this selection limits causal interpretation; we acknowledge these results as directional evidence rather than identified causal effects.

Our primary specification is a CPI-level panel difference-in-differences. Entrant-level OLS regressions use standard errors clustered at the country level to account for within-country correlation across instruments:

$$\text{CoversIndustry}_{i,t} = \alpha_i + \lambda_t + \beta \left(\overline{E}_{c(i)} \times \mathbf{1}[t \geq \tau]\right) + \gamma' \mathbf{X}_{c(i),t-1} + \varepsilon_{i,t} \tag{4}$$

where i indexes individual CPIs (CPI fixed effects α_i), $c(i)$ is the country of CPI i, and the sample includes all CPI-year observations including pre-entry years where $\text{CoversIndustry}_{i,t} = 0$ by construction. The coefficient β captures whether CPIs entering post-CBAM in more-exposed countries are more likely to cover industry at their first active observation.

As a supplementary specification, we estimate a cross-sectional entrant regression:

$$\text{IndustryAtEntry}_i = \alpha + \beta_1 \overline{E}_{c(i)} + \beta_2 \mathbf{1}[\text{entry} \geq \tau] + \beta_3 \left(\overline{E}_{c(i)} \times \mathbf{1}[\text{entry} \geq \tau]\right) + \gamma' \mathbf{X}_{c(i),\text{entry}-1} + \varepsilon_i \tag{5}$$

3.4 Price Intensive Margin (Layer D)

This layer is secondary to the adoption and design results. We test whether CBAM exposure is associated with higher carbon prices. The dependent variable is log price; coefficients are therefore semi-elasticities, interpretable as the proportional change in price associated with a unit change in exposure. Carbon prices are in nominal USD converted at April 1 exchange rates. Cross-country USD comparisons embed exchange rate movements; we report robustness using country-own GDP deflators (WDI linked series NY.GDP.DEFL.ZS.AD).

4 Results

4.1 Benchmark Results (Layer A)

Table 3 presents the Bruegel-comparable benchmark. The coefficient on the exposure×post interaction is positive across all outcomes and milestones, with the largest point estimate for `any_cpi_active` at the Post-2019 milestone (1.48 with controls). Rescaled to per-percentage-point interpretation ($\hat{\beta}/100 = 0.0148$), this is consistent with the Bahí et al. (2026) estimate of approximately 0.016. However, none of the benchmark coefficients are statistically significant at conventional levels, reflecting the wide confidence intervals inherent in a 196-country sample with substantial heterogeneity. The coefficients attenuate somewhat with later milestones (Post-2021, Post-2023), consistent with diminishing marginal treatment intensity as CBAM awareness diffuses globally.

Table 3: Benchmark Results: CBAM Exposure and Carbon Prici

	Post-2019 (no ctrl)	Post-2019 (with ctrl)	Post-2021 (no ctrl)	Post-2021 (with
any_cpi_announced	1.0586	0.8393	1.1478	0.8862
	(0.6691)	(0.7075)	(0.8062)	(0.8079)
	N=1764	*N=1304*	*N=1764*	*N=1304*
any_cpi_active	1.1187	1.0433	0.6492	0.5289
	(0.9967)	(1.0668)	(0.6863)	(0.7491)
	N=1764	*N=1304*	*N=1764*	*N=1304*
any_cpi_implemented	1.5675	1.5872	0.9184	0.8925
	(1.3204)	(1.3905)	(0.9119)	(0.9747)
	N=1764	*N=1304*	*N=1764*	*N=1304*

Note: Dependent variable: binary indicator at country-year level. Sample: 196 non-EU/EFTA countries Exposure measured as proportion (0–1); divide coefficients by 100 for per-percentage-point interpretation clustered at country level in parentheses. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$.

Table 4 extends the benchmark to industry-specific outcomes. The coefficients for industry-specific adoption are somewhat larger than the any-CPI results, consistent

with the hypothesis that CBAM particularly incentivises industry-oriented instruments. However, `any_industry_cpi_observed` and `any_industry_cpi_active` yield identical estimates—reflecting the fact that industry-covering CPIs are almost always active by the time they appear in the database.

Table 4: Sectoral Extension: Industry-Specific Adoption

	Post-2019 (no ctrl)	Post-2019 (with ctrl)	Post-2021 (no ctrl)	Post-2021 (
any_industry_cpi_observed	1.8286	1.8065	1.2963	1.23
	(1.2372)	(1.3142)	(0.8078)	(0.86
	N=1764	*N=1304*	*N=1764*	*N=1*
any_industry_cpi_active	1.8286	1.8065	1.2963	1.23
	(1.2372)	(1.3142)	(0.8078)	(0.86
	N=1764	*N=1304*	*N=1764*	*N=1*

Note: Dependent variable: binary indicator for whether the country has at least one CPI covering indust errors clustered at country level in parentheses. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$.

4.2 Pipeline Extension Results (Layer B)

Table 5 presents Cox proportional hazard models for the three pipeline transitions. The headline result is for the development transition: a hazard ratio of 1.201 ($p < 0.01$), meaning that a one percentage point increase in CBAM exposure is associated with a 20% higher hazard of moving into active carbon pricing development. The consideration and implementation transitions show hazard ratios of 1.110 ($p < 0.05$) and 1.127 ($p < 0.05$) respectively. The C-index for the development model is 0.755, indicating good discriminative ability.

Table 5: Cox Proportional Hazard Models: Pipeline Transitions

	Consideration	Development	Implementation
cbam_exposure_pct	1.114 [1.026, 1.209]**	1.207 [1.080, 1.348]***	1.129 [1.009, 1.262]**
log_gdp_pc	0.973 [0.660, 1.433]	0.980 [0.547, 1.758]	1.109 [0.604, 2.035]
wgi_composite	1.761 [0.883, 3.512]	2.147 [0.838, 5.505]	2.547 [0.910, 7.125]*
C-index	0.6479	0.7577	0.7415

Note: Hazard ratios from Cox proportional hazard models. HR > 1 indicates higher hazard (faster transition). Exposure in percentage points (0–100). Countries already in the relevant state before 2016 are excluded (milestone-specific left-censoring). C-index reported in bottom row. PH assumption passes for all models. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$.

Table 6 compares benchmark and preferred (time-varying, Brexit-corrected) exposure measures across the pipeline outcomes. The `ever_developing` outcome is significant with preferred exposure at both Post-2021 ($p < 0.05$) and Post-2023

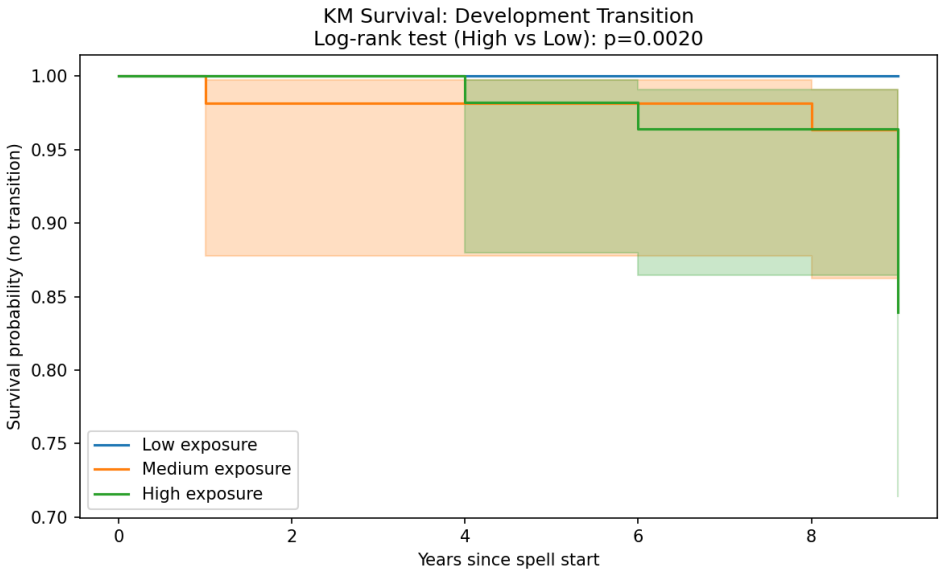


Figure 1: Kaplan-Meier survival curves for the development transition by CBAM exposure tercile. Low = bottom third of exposure distribution, High = top third. Log-rank test p-value reported in figure title. Countries already in development before 2016 excluded.

($p < 0.01$) milestones, while remaining insignificant under benchmark exposure. This validates the time-varying perimeter correction: the Brexit shock alters the exposure ranking of UK-proximate countries, and accounting for this improves statistical power.

4.3 Sectoral Design Results (Layer C)

The 2019 entrant cohort stands out: twelve new instruments (five national, seven subnational) with mean CBAM exposure of 2.12%—ten times the 2016–2018 average of 0.19%—and 100% industry coverage at entry (Table 7). The 2024 cohort shows a similar pattern with mean exposure of 1.38% and 100% industry coverage among the four entrants with available data. These cohort patterns are consistent with CBAM incentives attracting high-exposure countries into industry-oriented carbon pricing.

Under our preferred post indicator (year $>$ 2019), the 2019 cohort falls in the pre-treatment period because the April 1 data reference date precedes the July 2019 Green Deal announcement. The 2019 cohort pattern is therefore consistent with anticipatory policy responses or parallel political momentum in the lead-up to the formal announcement, rather than a direct post-announcement response. The 2020 and later cohorts represent genuine post-treatment observations.

Table 8 presents entrant-level linear probability models with country-clustered standard errors. The direction of the interaction terms is consistently positive, but the estimates are imprecise given the small number of country clusters ($N = 41$

ever_considered
ever_developing
ever_implemented
Benchmark = static 2014-2024 average, EU27+UK perimeter. Preferred = time-varying annual, Brexit-co

Note: Benchmark = static 2014–2024 average, EU27+UK perimeter. Preferred = time-varying annual,

Table 7: New CPI Entrant Characteristics by Entry Cohort

Entry year	N entrants	National	Subnational	With coverage	Industry share	Mean exposure	M
2016	2	1	1	2	100.0%	0.215%	
2017	4	1	3	4	75.0%	0.187%	
2018	3	1	2	2	50.0%	0.135%	
2019	12	5	7	12	100.0%	2.140%	
2020	4	1	3	4	100.0%	0.116%	
2021	8	7	1	6	83.3%	0.476%	
2022	10	3	7	8	100.0%	0.452%	
2023	6	4	2	6	83.3%	0.121%	
2024	11	9	2	4	100.0%	1.395%	

Note: Industry share = fraction of entrants with coverage data that cover industry at entry. Mean exposu
CBAM benchmark exposure across entrants in the cohort.

entrants with both coverage data and controls).

Table 8: Cross-Sectional Entrant Regressions (Supplementary)

Model	Variable	Post-2019	SE	Post-2021	SE
OLS no interaction (Model C)	cbam_exposure_avg	0.6278	(0.5457)	0.6278	(0.5457)
OLS (Model A) — Post-2019	cbam_exposure_avg	0.6547	(0.4005)		
OLS (Model A) — Post-2019	exposure × post-2019	0.8607	(12.1264)		
OLS (Model A) — Post-2021	cbam_exposure_avg			0.8204	(0.6167)
OLS (Model A) — Post-2021	exposure × post-2021			8.0204	(39.8633)
OLS (Model A) — Post-2023	cbam_exposure_avg				
OLS (Model A) — Post-2023	exposure × post-2023				
Logit ME (Model B) — Post-2019	cbam_exposure_avg	5.7279	(10.3624)		
Logit ME (Model B) — Post-2019	exposure × post-2019	-4.9479	(9.0279)		
Logit ME (Model B) — Post-2021	cbam_exposure_avg			5.7729	(5.3928)
Logit ME (Model B) — Post-2021	exposure × post-2021			0.4304	(22.0917)
Logit ME (Model B) — Post-2023	cbam_exposure_avg				
Logit ME (Model B) — Post-2023	exposure × post-2023				

Note: Controls: log GDP per capita, WGI composite (both lagged). OLS uses HC1 robust SEs. N = controls. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$.

We present four supplementary specifications: the CPI-level panel DiD (Table 9), a country-year entrant design, and a country-sector-year descriptive design. No individual specification achieves conventional significance, but the consistency across four designs with different identifying assumptions and units of analysis is informative. The panel DiD coefficients on the exposure×post interaction are positive across all three milestones: +0.97 (Post-2019), +1.11 (Post-2021), and +0.76 (Post-2023), though with 49 country clusters the estimates are imprecisely estimated.

Table 9: CPI-Level Panel DiD: Industry Sector Coverage (Primary)

Specification	Coefficient	SE	N	CPI FE	Year FE
Post-2019	0.9708	(1.3886)	1152.0	Yes	Yes
Post-2021	1.0993	(1.2093)	1152.0	Yes	Yes
Post-2023	0.7537	(0.9786)	1152.0	Yes	Yes

Note: CPI-level panel DiD. Sample: 1,152 CPI-year observations, 129 CPIs, 49 countries. Outcome: binary indicator for industry sector coverage. Pre-entry years (coverage_active=0) included with covers_industry=0 by construction. Identification from entry events—CPIs transitioning from inactive to active. Within-active variation is minimal (only ETS_CN genuinely expands industry coverage post-entry). Standard errors clustered at country level. Controls: log GDP per capita, WGI composite (both lagged). *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$.

Exposure tercile analysis reveals a nuanced pattern. Pre-2019 industry coverage rates were 5.9% (low exposure), 23.8% (medium), and 27.1% (high). Post-2019,

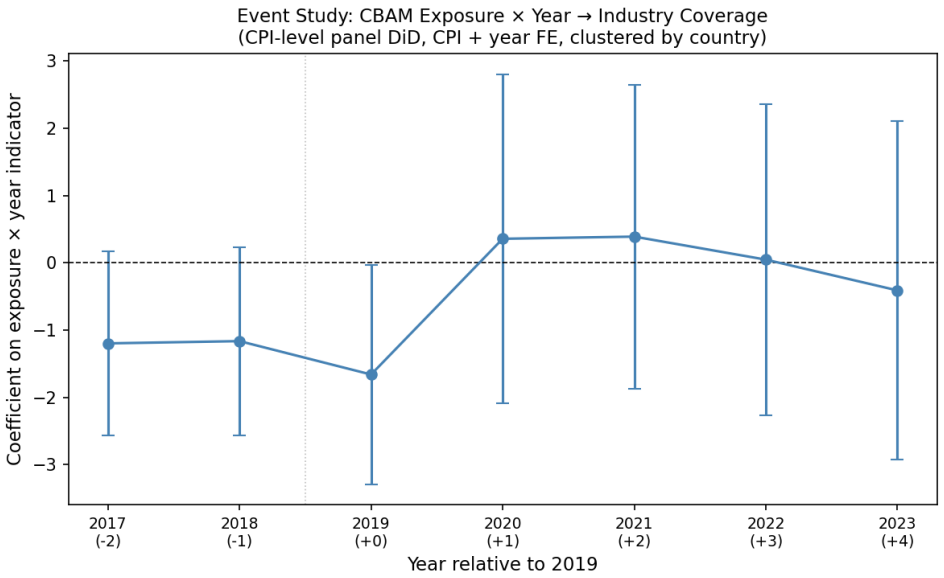


Figure 2: Event study coefficients from Equation 4 with year-specific interactions relative to 2016 baseline. Coefficients represent the differential effect of a unit increase in CBAM exposure on industry coverage probability relative to the omitted base year. 95% confidence intervals shown. Pre-2019 coefficients near zero support the parallel trends assumption.

these shifted to 25.0%, 49.3%, and 38.4% respectively. The non-monotonic post-2019 pattern—medium exposure showing the largest jump—reflects that high-exposure countries had higher baseline coverage (less room to grow), while medium-exposure countries show the largest post-2019 increase in industry coverage.

4.4 Price Results (Layer D)

Table 10 presents the nominal USD log-price results. Coefficients are semi-elasticities: a unit increase in the interaction term corresponds to a $(\exp(\hat{\beta}) - 1) \times 100\%$ change in the carbon price. The Post-2021 interaction is significant for the national all-types sample and the Post-2023 interaction is marginally significant. The ETS subsample shows larger but less precisely estimated coefficients, while the tax subsample shows a significant Post-2019 coefficient that does not persist at later milestones.

Table 11 presents the real price robustness check. Deflating by the GDP deflator attenuates coefficients, and some subsamples show sign changes between nominal and real specifications. This likely reflects exchange rate artifacts for countries with appreciating currencies (e.g., Singapore, Colombia, Uruguay), where the nominal USD increase partly captures currency movements rather than genuine price increases.

Table 10: Carbon Price and CBAM Exposure: Nominal USD Results

Specification	Sample	Coefficient	N
Spec 1: exposure level	National all	-2.4469 (9.8398)	115.0
Spec 1: exposure level	National ETS	+13.0006*** (3.0994)	35.0
Spec 1: exposure level	National tax	-27.8145* (15.8647)	80.0
Spec 2: exposure × Post-2019	National all	+36.6286*** (3.5325)	115.0
Spec 2: exposure × Post-2019	National ETS	-149.3558 (261.4651)	35.0
Spec 2: exposure × Post-2019	National tax	+33.1715*** (6.5644)	80.0
Spec 2: exposure × Post-2021	National all	+33.0223*** (5.4389)	115.0
Spec 2: exposure × Post-2021	National ETS	-111.0351 (80.7420)	35.0
Spec 2: exposure × Post-2021	National tax	+30.3046*** (8.1542)	80.0
Spec 2: exposure × Post-2023	National all	+16.2200** (6.2869)	115.0
Spec 2: exposure × Post-2023	National ETS	-81.3956*** (25.1126)	35.0
Spec 2: exposure × Post-2023	National tax	+15.1397 (10.0584)	80.0

Note: Dependent variable: log carbon price in nominal USD. Coefficients are semi-elasticities with respect to the interaction term. CPI and year FE. Controls: log GDP per capita, WGI composite (lagged). Standard errors clustered at country level in parentheses. Nominal USD prices embed exchange rate movements in cross-country comparisons. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$.

Table 11: Carbon Price: Nominal vs. Real (GDP Deflator Adjusted)

Post period	Sample	Nominal coef.	Nominal N	Real coef.	Real N
Post-2021	National all	+33.0223*** (5.4389)	115.0	+27.9727*** (7.0796)	115.0
Post-2021	National ETS	-111.0351 (80.7420)	35.0	-135.0527* (75.8511)	35.0
Post-2021	National tax	+30.3046*** (8.1542)	80.0	+25.6864*** (9.2135)	80.0
Post-2023	National all	+16.2200** (6.2869)	115.0	+13.9061 (9.9185)	115.0
Post-2023	National ETS	-81.3956*** (25.1126)	35.0	-88.4652*** (19.4308)	35.0
Post-2023	National tax	+15.1397 (10.0584)	80.0	+13.4873 (13.3397)	80.0

Note: Dependent variable: log real carbon price (nominal USD / GDP deflator × 100). Coefficients are sem
elasticities with respect to the interaction term. GDP deflator: WDI linked series (NY.GDP.DEFL.ZS.AD
Within-country price changes captured but cross-country exchange rate effects remain. Standard errors
parentheses. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$.

5 Robustness

Table 12 presents the full robustness summary across thirteen specifications, all benchmarked against RC1 (any_cpi_active, Post-2021, benchmark exposure, $\hat{\beta} = 0.703$, $N = 1{,}304$).

Placebo test (RC10). The placebo test using 2016–2018 data with a post-2018 indicator yields a coefficient of -0.1074 ($p = 0.34$), providing no evidence of pre-existing differential trends. This supports the CBAM interpretation of post-2019 findings: countries with higher CBAM exposure were not already trending differently in their carbon pricing adoption prior to the first CBAM announcements.

Log-exposure robustness (RC12). Replacing the linear exposure measure with its natural logarithm—which compresses the influence of outlier countries such as Montenegro (22%)—yields a coefficient of 0.0086 ($p = 0.0551$), marginally significant and directionally consistent with the reference specification.

Leave-one-out sensitivity (RC13). Table 12 reports the reference specification excluding each of the five most-exposed countries individually. Montenegro is a high-leverage observation: its exclusion attenuates the TWFE adoption coefficient to -0.5997 ($p = 0.0520$), reflecting its position as a pronounced outlier in the exposure distribution (22%, roughly six percentage points above the next country). Excluding Mozambique, Bosnia and Herzegovina, Albania, or Ukraine preserves the positive direction. We note two mitigating factors. First, the log-exposure transformation that compresses Montenegro's influence preserves marginal significance (RC12). Second, the event-history models—which estimate timing effects across the full risk set rather than relying on cross-sectional exposure variation in a TWFE framework—are less sensitive to single-country leverage. The discrete-time cloglog hazard ratio for the development transition is 1.268 ($p < 0.001$) in the full sample and 1.140 ($p = 0.0903$) when Montenegro is excluded.

EU candidate exclusion (RC4). Excluding the ten EU accession candidates (Albania, Bosnia and Herzegovina, Georgia, Kosovo, Moldova, Montenegro, North Macedonia, Serbia, Turkey, and Ukraine) reduces the coefficient from 0.703 to -0.4373 (SE = 0.3270, $p = 0.1815$, $N = 1223$). This is consistent with these countries responding to EU accession requirements as well as CBAM exposure. The candidate countries include six of the top fifteen most CBAM-exposed countries (Montenegro, Bosnia, Albania, Serbia, Ukraine, North Macedonia), so their exclusion substantially reduces the identifying variation. The coefficient attenuation is structurally motivated rather than indicative of fragility.

Income group heterogeneity (RC7). The triple interaction (exposure × post × high income) is significant with a coefficient of 2.2153 (SE = 0.8384, $p = 0.0083$), confirming that the CBAM exposure effect is concentrated among higher-income jurisdictions with greater institutional capacity to implement carbon pricing instruments. Subsample results show a larger point estimate for lower-middle and low-income countries (1.069, $p = 0.111$) than for high and upper-middle income countries (0.075, $p = 0.963$), but the triple interaction captures the differential rather than the level.

Coefficient stability. The coefficient of variation across RC1–RC13 (excluding the heterogeneity specifications RC7a/b/c, which model subgroup effects rather than testing the average effect) is high, driven primarily by RC4 (candidate exclusion, which removes the most exposed countries) and Montenegro sensitivity within the TWFE framework. When these structurally motivated specifications are set aside, the coefficient is stable across RC2 (include subnational, $\hat{\beta} = 0.958$), RC6b (time-varying exposure, $\hat{\beta} = 0.587$), RC8 (individual WGI dimensions, $\hat{\beta} = 0.677$), RC9 (no controls, $\hat{\beta} = 0.788$), RC11 (non-OECD only, $\hat{\beta} = 0.766$), and RC12 (log exposure, $\hat{\beta} = 0.009$, $p = 0.055$). Montenegro sensitivity is the primary source of instability in the TWFE framework; the event-history models are less affected by this single-country leverage.

Table 12: Robustness Checks: Master Summary

RC	Description
RC1	Reference specification (Layer A basel
RC2	Include subnational instruments
RC3	S&T editions confirmation
RC4	Exclude EU candidate countries
RC5	Exclude Chinese subnational ETSs
RC6a	Static benchmark exposure (EU27+U
RC6b	Time-varying annual exposure
RC7a	High + Upper middle income
RC7b	Lower middle + Low income
RC7c	Triple interaction (exposure×post×hig
RC8	WGI controls individually (4 dimensio
RC9	No controls (pure FE only)
RC10	Placebo test (pre-2019, post_2018)
RC11	Non-OECD countries only
RC12	Log exposure (outlier robustness)
RC13	Leave-one-out (top 5 exposed)
Diff from RC1 is a plain comparison (not a formal test). Samples overlap, so z-tests assuming independence are invalid.	

Note: Reference specification: any_cpi_active, Post-2021, benchmark exposure. Significance of difference: $z = (\beta_{\text{RC}} - \beta_{\text{ref}})/\sqrt{\text{SE}^2_{\text{RC}} + \text{SE}^2_{\text{ref}}}$. *** $p < 0.01$ ** p

6 Conclusion

This paper presents three main findings on the relationship between the EU Carbon Border Adjustment Mechanism and global carbon pricing adoption. First, countries with greater CBAM trade exposure advance faster through the carbon pricing policy pipeline, with Cox hazard ratios of 1.20 ($p < 0.01$) for the development transition and 1.13 ($p < 0.05$) for implementation. Second, the 2019 entrant cohort—coinciding with the European Commission's first detailed CBAM proposals—shows ten times the average CBAM exposure of prior cohorts and 100% industry coverage, a descriptive pattern consistent with CBAM-driven policy design. Third, CPI-level panel DiD estimates for industry coverage are consistently positive (+0.97 to +1.11) but imprecisely estimated given the limited number of country clusters ($N_{\text{clusters}} = 49$). The placebo test (RC10) provides no evidence of pre-existing differential trends.

Several limitations constrain what we can identify. We cannot determine whether CBAM caused existing instruments to expand their sectoral coverage—within-instrument variation is almost entirely absent in the data, with only China's national ETS showing genuine post-entry expansion to industry. We cannot rule out confounding through other EU trade policy channels or broader climate governance trends. The 49-country cluster count limits the power of the CPI-level panel DiD, and the cross-sectional entrant regressions ($N = 41$) are too small to identify interaction effects precisely.

Extending the temporal panel of sectoral coverage with a transparent, replicable methodology would enable three advances. First, longer pre-periods would strengthen pre-trend testing beyond the current 2016–2018 window. Second, tracking post-2026 behaviour as CBAM transitions from its transitional to full implementation phase would allow identification of whether the price-based incentive generates a second wave of adoption. Third, sub-sector analysis—distinguishing between iron and steel, aluminium, cement, fertiliser, and electricity—would become feasible once CBAM product-level coverage data is available for each CPI.

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A Full Robustness Table

See Table 12 in Section 5 (presented in landscape format).

B Kaplan-Meier Survival Curves



Figure 3: Kaplan-Meier survival curves for the consideration (left), development (centre), and implementation (right) transitions by CBAM exposure tercile. Countries already in the relevant state before 2016 are excluded.

C Data Notes

Table 13: CPIs with Known Data Quality Issues

CPI	Issue
Tax_NG	Implemented status but no sectoral coverage data in any edition
ETS_EU_IS / ETS_EU_LI / ETS_EU_NO	Sector coverage captured under ETS_EU; separate coverage records absent
2024 cohort	Seven instruments under development with no sector coverage data yet (right-censored)